

**1****1942**

A theropod skull was found in Hell Creek Formation rocks from Montana. This formation is known for producing *Tyrannosaurus rex* specimens, and while the skull closely resembles a *T. rex*, it is only about half the length.

**2****1946**

The 1942 skull resembles *Gorgosaurus*, a tyrannosaur from Alberta, in size and proportions. *Gorgosaurus* lived about 9 million years before the Hell Creek rocks formed.

**3****1965**

Some small tyrannosaur skeletons from Mongolia, once thought to be separate species, are young *Tarbosaurus*.

**4****1988**

The bones of the 1942 skull appear fused to one another.

**5****1988**

The 1942 skull has 15 teeth in one side of its upper jaw. Adult *T. rex* have 11 or 12.

**6****1999**

The joints between the bones of the 1942 skull are open, not fused.

**7****1999**

In *Gorgosaurus*, young animals have more teeth than adults: tooth number falls as the animal grows.

**8****2001**

Jane, a second small tyrannosaur from Hell Creek, is more than half a skeleton with a nearly complete skull.

**9****2003**

In Albertosaurus, Gorgosaurus and Daspletosaurus, young animals have low, narrow skulls and adults have deep, heavy skulls.

**10****2011**

A juvenile Tarbosaurus has about the same number of teeth as adult Tarbosaurus.

**11****2013**

The Dueling skeleton, a third small tyrannosaur from Hell Creek, has hand proportions and a wishbone (furcula) shape that differ from T. rex.

**12****2020**

Bone sections from Jane show growth rings and bone tissue typical of an animal that was still growing.

**13****2024**

In a family-tree analysis of anatomical characters, the 1942 skull and Jane may fall outside Tyrannosauridae, the family that contains T. rex.

**14****Jun 2025**

In tyrannosaur growth series, many characters used to build family trees differ between young and adult animals of one species.

**15****Oct 2025**

The Dueling skeleton's skull is 71 centimeters long, longer than the 1942 skull's 57.

**16****Oct 2025**

The outermost growth rings in the Dueling skeleton's bone are tightly packed, a pattern seen when growth has nearly stopped.

**17****Oct 2025**

Growth rings put the Dueling skeleton at 17 to 22 years old at death, and Jane at 8 to 14.

**18****Oct 2025**

The Dueling skeleton has 16 teeth in the left side of its upper jaw and 17 in the right. Adult T. rex have 11 or 12.

**19****Oct 2025**

The arm and hand bones of the Dueling skeleton are proportionally larger than those of T. rex.

**20****Oct 2025**

In living and fossil land vertebrates, limb bones do not shrink in absolute size as the animal grows.

**21****Oct 2025**

The Dueling skeleton has fewer tail vertebrae than T. rex.

**22****Oct 2025**

The Dueling skeleton has several air-sac openings (pleurocoels) in its second neck vertebra and in its tail vertebrae. Jane has one in the second neck vertebra and none in the tail.

**23****Dec 2025**

Bone sections from the throat bones (hyoids) of the 1942 skull show it was a mature animal.

**24****Jan 2026**

Growth patterns in Jane's bone do not match the growth curve built from T. rex hindlimbs of many sizes.